
Euclidean Geodesic Alignment: Mitigating the ID-OOD Tradeoff in Adapter Tuning for Retrieval

Anonymous Author(s)

Affiliation

Address

email

Abstract

We study how lightweight adapters trained on top of frozen vision-language encoders behave under distribution shift, and identify a structural in-distribution/out-of-distribution (ID/OOD) tradeoff among existing adapter training paradigms. Global contrastive objectives such as ICon and SRL, when paired with high-capacity adapters, achieve near-perfect in-distribution Label Precision on CIFAR-100 (0.999 and 0.992 at $K = 1$, $n_{\text{probe}} = 1$) but collapse to 0.557 and 0.520 on the unseen classes of Food-101: a 44–47 percentage point inversion between the two scenarios. Low-capacity alternatives such as LoRA avoid this collapse but underperform substantially in-distribution. We propose **Euclidean Geodesic Alignment (EGA)**, a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. EGA converges to a sparse-gradient scheme in which 96.5% of training triplets produce zero gradient at steady state, preserving the pre-trained semantic structure for unseen categories while still permitting high-capacity in-distribution refinement. Across CIFAR-100 (ID) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), the six methods we evaluate trace out a Pareto frontier in the (ID, OOD) Label Precision plane, and EGA is the only adapter that is not strictly dominated: methods that exceed EGA’s in-distribution precision (ICoN, SRL) collapse on OOD; methods that match EGA on OOD (LoRA, frozen baseline) underperform in-distribution.

1 Introduction

Modern retrieval systems increasingly rely on embedding models such as CLIP [24] for their remarkable zero-shot semantic capabilities. These models provide highly generalizable latent spaces that serve as excellent starting points for vector similarity search. However, when deploying these systems to specialized domains, practitioners often seek to further align the embeddings with domain-specific category structures to maximize retrieval accuracy. Because full-scale fine-tuning is computationally prohibitive and risks destroying the foundational zero-shot knowledge, parameter-efficient adaptation has emerged as the standard paradigm: lightweight adapters are trained on a labeled subset of the target task while the massive pre-trained backbone remains frozen.

A subtle but consequential aspect of this paradigm is that retrieval systems do not only serve queries from the categories seen during adapter training. In real deployments, the database and the query stream routinely contain *unseen* categories that were absent from the labeled subset, and the adapter must continue to support meaningful retrieval over these. The standard practice is training adapters with global contrastive objectives such as ICon [1] and SRL [4] that compute losses over the full pairwise similarity matrix in each mini-batch; however, this approach optimizes aggressively for the seen-class structure with no explicit mechanism to preserve the embedding geometry of unseen categories.

We show that this paradigm produces a structural in-distribution/out-of-distribution (ID/OOD) tradeoff rather than a uniformly improved adapter. On in-distribution data (CIFAR-100), high-capacity global contrastive adapters reach Label Precision near 1.0: ICon at 0.999, SRL at 0.992 at $K = 1$, $n_{\text{probe}} = 1$. On the strictly unseen classes of Food-101, the same adapters collapse to 0.557 and 0.520, dropping 44 and 47 percentage points below their own in-distribution scores and 32 to 36 percentage points below the *frozen* CLIP baseline. Restricting capacity through low-rank adapters such as LoRA [10] avoids this catastrophic collapse but pays a different cost: LoRA’s in-distribution Label Precision is bounded around 0.61–0.67, far below what high-capacity adapters can achieve when overfitting is not punished by distribution shift.

We propose Euclidean Geodesic Alignment (EGA), a manifold-preserving adapter that occupies a Pareto-optimal balance between these schemes. EGA combines three design choices: (i) a zero-initialized residual structure, (ii) an ℓ_2 normalization constraint, and (iii) a small-margin triplet loss on the class-aware neighborhood graph. The triplet loss with small margin produces a self-limiting optimization dynamic: as the adapter converges on seen classes, the fraction of triplets violating the margin decays rapidly, and at steady state 96.5% of training triplets contribute zero gradient. The high-capacity architecture is therefore allowed to tightly refine the regions of the embedding space that the seen classes occupy, while remaining largely inactive over the rest, which is a property that protects the pre-trained semantic structure of unseen categories from the dense global reshaping that drives ICon and SRL to OOD failure.

We evaluate EGA on CIFAR-100 (in-distribution) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), comparing against the frozen CLIP baseline, global contrastive adapters (ICoN, SRL), and two LoRA configurations (LoRA+InfoNCE, LoRA+Triplet). The six methods trace out a Pareto frontier in the (ID, OOD) Label Precision plane, and EGA is the only adapter that is not strictly dominated by any other: it reaches 0.851 in-distribution Label Precision (within 15 percentage points of the ICoN/SRL ceiling), while improving over the frozen baseline on Aircraft (+3.6 pp) and CIFAR-10 (+0.6 pp) and bounding the degradation on Food-101 (−8.3 pp, versus −32 to −36 pp for ICoN and SRL).

2 Related Work

Manifold Preservation under Adapter Tuning Foundation models such as CLIP [24] produce embedding spaces whose structure encodes both seen class semantics and a broader prior over visual concepts not observed during downstream training. Catastrophic forgetting has been studied extensively in continual and incremental learning settings [20, 36]. Recent work has documented analogous degradation phenomena specific to contrastive representations: anisotropic collapse into narrow conic regions [41, 6], structural fracturing under modality gap [25, 5], and loss of compositional structure under aggressive optimization [23, 27, 35]. To counter these degradations, recent adaptation strategies impose explicit geometric constraints such as orthogonality regularization [26] or restrict optimization to limited subspaces. EGA leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen class regions of the pretrained manifold structurally unmodified.

Global Contrastive Objectives and Structural Regularization Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [30, 19]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution [33], a strategy applied across various domains including graph encoders [32] and multimodal recommendation systems [42]. While pushing embeddings toward a uniform distribution enhances robustness to background noise [34], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like ICoN formulate a unifying framework to systematically align multi view representations [1], while geometric constraints are leveraged to improve imbalanced regression tasks [4]. Other approaches attempt to decouple the learning objectives [18] or apply information theoretic regularizers [40] to control the density of the learned representations. Furthermore, region based distillation techniques have been proposed to enhance fine grained understanding [15]. EGA replaces dense global optimization with sparse local triplet updates, preserving out of distribution semantic structure.

Parameter-Efficient Manifold Adaptation Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [9, 10]. This paradigm has extensively shifted towards vision language models, dominating recent literature due to the prohibitive cost of full scale fine tuning [21]. Methods such as Visual Prompt Tuning [14] and Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [39]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine tuning while preserving the generalizable representations of the foundation model [28]. To further protect the pre trained manifold during initial training steps, zero initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [38]. EGA integrates a zero initialized residual adapter with a localized geometric objective, achieving both parameter efficiency and targeted manifold smoothing.

Vector Similarity Search and Metric Consistency Vector similarity search is the foundational engine for retrieving high dimensional representations at scale. The systems community has developed highly optimized indexing algorithms to handle billion scale datasets. Foundational works such as FAISS leverage hardware acceleration for fast similarity search [16], and Hierarchical Navigable Small World graphs establish the core topology for efficient approximate retrieval [22]. Meanwhile, hybrid structures like DiskANN and SPANN achieve exceptional speed by efficiently managing memory and disk hierarchies [29, 2]. Recent advancements continue to refine these topologies, exploring crossing sparse proximity graphs [37] and analyzing the theoretical constructions of navigable graphs for high dimensional nearest neighbor search [3]. Furthermore, system level studies optimize retrieval latency and stability by aligning best first search algorithms with solid state drives [7] and enabling direct insertions in large scale indices [8]. To reduce the computational cost of exact distance calculations during the retrieval phase, approximation techniques such as low rank matrix factorization have also been introduced to accelerate inner product evaluations [13]. However, extensive theoretical analyses reveal that popular approximate nearest neighbor implementations still face severe worst case performance limitations [12]. EGA proactively recalibrates the metric manifold prior to index construction, enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$.

Our goal is to learn a lightweight adapter $f_\theta : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

Evaluation metrics. We evaluate retrieval quality along two complementary dimensions.

Label Precision@ K measures the semantic quality of retrieved results—whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^K \mathbf{1}[y_{\hat{n}_k(q)} = y_q], \quad (1)$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k -th nearest neighbor of q in the transformed embedding space.

ANNS Recall@ K measures the geometric fidelity of the ANN index—the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget

137 nprobe:

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)|}{K}, \quad (2)$$

138 where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top- K neighbor sets, respectively.

139 The two metrics capture distinct failure modes: an adapter may improve LP@ K by tightening
140 within-class clusters (semantic improvement) while simultaneously improving AR@ K by making the
141 geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4,
142 global contrastive objectives achieve high AR@ K on seen classes but collapse on *both* metrics for
143 unseen classes, whereas EGA improves both simultaneously under the OOD setting.

144 3.2 Manifold-Preserving Adapter Design

145 EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement
146 layer. The architecture is deliberately constrained by three design principles, each of which serves a
147 distinct role in preserving the pretrained manifold.

148 **Residual connection with zero initialization.** Each EGA block applies a residual update of the
149 form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \quad (3)$$

150 where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely,
151 the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This
152 guarantees that training begins from the pretrained geometry rather than a random perturbation, and
153 that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation
154 (Table 3), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because
155 the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

156 **Feature gating.** Inside each residual block, the transformed signal passes through a feature-gating
157 module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2 \text{GELU}(W_1 \mathbf{h})), \quad (4)$$

158 where $W_1 \in \mathbb{R}^{(d/4) \times d}$, $W_2 \in \mathbb{R}^{d \times (d/4)}$, σ is the sigmoid function, and $\odot$ denotes elementwise
159 multiplication. This structure, analogous to Squeeze-and-Excitation networks [11], learns a per-
160 dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones.
161 The bottleneck ratio of 1/4 is chosen to prevent the gating network from overfitting on seen-class
162 data, as validated in Table 4.

163 **ℓ_2 normalization.** The final output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2
164 normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \quad (5)$$

165 This constraint ensures that the transformed embeddings remain in the same metric space as the
166 original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable
167 on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage
168 points (Table 3), confirming that hypersphere consistency is necessary for stable ANN indexing.

169 **Full forward pass.** Let Block_i denote the i -th EGA block consisting of a two-layer MLP with Lay-
170 erNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm.
171 The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z})))) . \quad (6)$$

172 The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension
173 2048, which is negligible compared to the frozen CLIP backbone ($\sim 150M$ parameters).

3.3 Local Triplet Training and Gradient Sparsity

Training objective. Given a mini-batch sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a is an anchor image, p is a positive sampled uniformly from the same class as a , and n is a negative sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \max(0, d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m), \quad (7)$$

where $d(\cdot, \cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the transformed embedding. We set $m = 0.2$ throughout all experiments based on the sensitivity analysis in Table 4.

Gradient sparsity and manifold preservation. A triplet (a, p, n) contributes a non-zero gradient to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \quad (8)$$

We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 5 (left) shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast majority of local neighborhood relationships in the embedding space.

This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided that the adapter’s updates remain effectively local (i.e., the cumulative perturbation induced by seen-class training gradients remains small in unseen-class regions of the embedding space), the CLIP manifold structure for unseen classes is largely preserved. The gradient sparsity mechanism enforces exactly this locality: once seen-class neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of actively-violated local relationships continue to be updated.

Contrast with global contrastive objectives. Global objectives such as ICon [1] and SRL [4] compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a KL divergence between the empirical similarity distribution and the class-membership distribution, while SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives generate dense gradient fields in every training step: every sample pair (regardless of whether its local geometry already satisfies the retrieval objective) contributes to the loss and receives a non-zero gradient. This global optimization pressure from seen-class training propagates throughout the embedding space, reshaping the relative positions of unseen-class regions according to seen-class similarities rather than preserving the pretrained semantic structure for those categories.

Proposition 1 (Gradient Sparsity at Convergence). *Let f_θ be trained with the triplet loss in Eq. (7) with margin $m > 0$ on a dataset $\mathcal{D}_{\text{train}}$ drawn from seen classes $\mathcal{Y}_{\text{seen}}$. Define the active triplet ratio at training step t as*

$$\rho(t) = \Pr_{(a,p,n) \sim \mathcal{B}} \left[d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0 \right]. \quad (9)$$

As f_θ converges, $\rho(t) \rightarrow \rho^*$, where

$$\rho^* = \Pr_{(a,p,n)} \left[d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0 \right] \quad (10)$$

is determined solely by the margin m and the within-class distance distribution of the converged embeddings. Furthermore, $\nabla_\theta \mathcal{L} = \mathbf{0}$ for all triplets (a, p, n) with $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$, so the adapter gradient is supported only on the fraction ρ^* of the training distribution. For embeddings whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.

Proof Sketch. The triplet loss consists of hinge terms whose gradient is zero whenever the margin constraint is satisfied. At convergence, only those triplets whose geometry the adapter has not yet separated within the margin remain active. The fraction of such triplets is determined solely by the margin m and the distance distribution of seen classes, independent of unseen classes. Since no training triplet involves unseen-class embeddings, the adapter exerts zero gradient over the entire unseen-class region of the embedding space. See Appendix A for the complete proof. $\square$

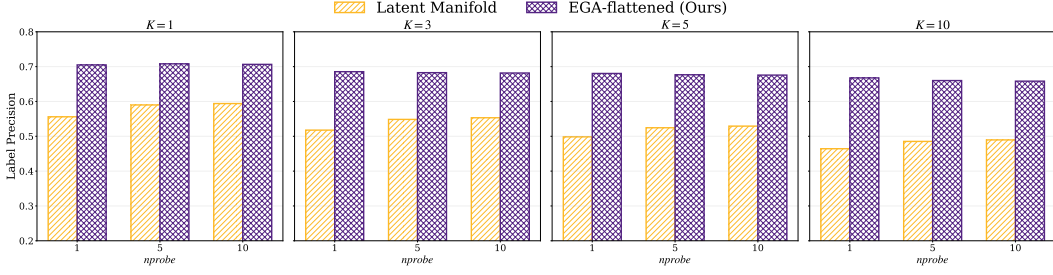


Figure 1: Label Precision@ K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and $nprobe \in \{1, 5, 10\}$.

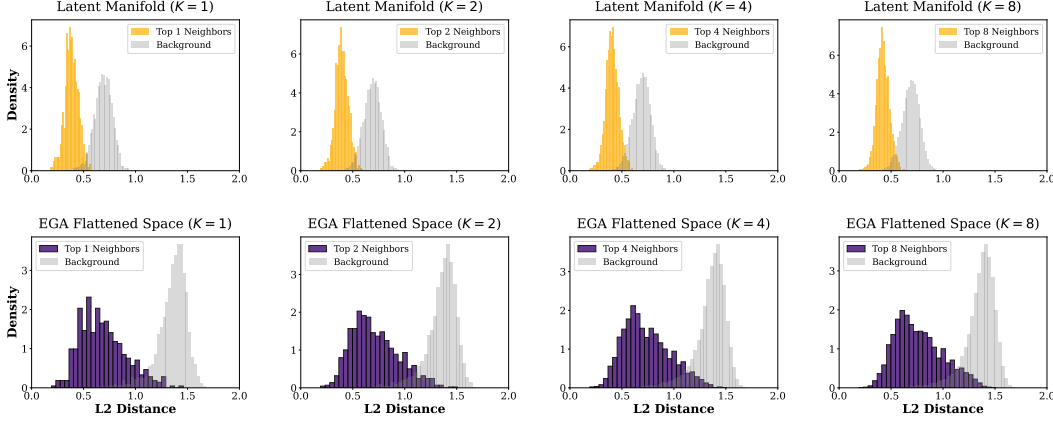


Figure 2: L_2 distance distributions for Top- K neighbors versus random background points on CIFAR-100.

4 Evaluation

Experiments were conducted on the Chameleon Cloud platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Evaluation datasets include CIFAR-100 for in distribution scenarios, CIFAR-10 for cross dataset OOD settings, and FGVC-Aircraft and Food-101 for unseen class OOD evaluations. All of these datasets can be directly downloaded from the Pytorch library. We compare our method against four baseline models: ICon [1], SRL [4], LoRA + InfoNCE [10, 31], and LoRA + Triplet [10].

Extended experimental details can be found in Appendix B.

4.1 In-Distribution Validation

Semantic retrieval quality. Figure 1 reports Label Precision@ K across four neighborhood sizes and three search budgets. EGA improves Label Precision@1 from 0.549 (raw CLIP) to 0.851 at $nprobe = 1$, and the gap persists across all K and $nprobe$ values. The improvement is not an artifact of larger search budgets: even at $nprobe = 10$, EGA retains a clear advantage: the adapter tightens within-class neighborhoods rather than benefiting from broader index coverage.

Geometric structure. The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance distributions for true Top- K neighbors against random background points, before and after EGA. In the raw CLIP space, the two distributions overlap substantially: the distance to a true neighbor is often indistinguishable from the distance to an unrelated point, which is exactly the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all K , with neighbors concentrated at small distances and background points pushed toward a distinct mode. This separation is the mechanism by which EGA improves the recall-efficiency frontier shown in Figure 3: the raw CLIP space achieves only 0.805 Recall@1 at $nprobe = 1$, whereas EGA reaches 0.933 and saturates above 0.97 by $nprobe = 5$.

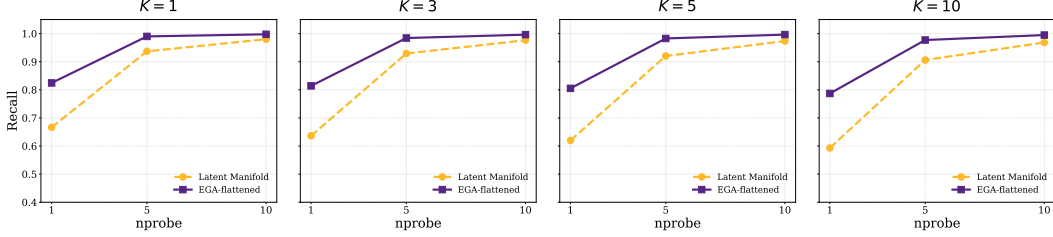


Figure 3: ANNS Recall@ K versus nprobe on CIFAR-100.

Table 1: In-distribution retrieval performance on CIFAR-100 at $K = 1$, nprobe = 1.

Metric	CLIP	ICo	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
LP@1	0.549	0.999	0.992	0.668	0.612	0.851
AR@1	0.805	0.992	0.991	0.879	0.908	0.933

In-distribution comparison across adapters. While Figures 1–3 focus on the EGA-vs-frozen contrast that underlies our geometric story, we also report all six methods’ in-distribution performance in Table 1, since the contrast with their out-of-distribution behavior (Section 4.2) is central to our analysis. ICo and SRL achieve near-perfect Label Precision (0.999 and 0.992 respectively) by maximally tightening seen-class clusters with their global contrastive objectives. EGA reaches LP@1 of 0.700, an improvement of +15 pp over the frozen baseline; the LoRA variants land between EGA and the baseline. As we show in Section 4.2, the same property that allows ICo and SRL to dominate this in-distribution metric is what causes their catastrophic out-of-distribution collapse.

4.2 Out-of-Distribution Generalization

We now turn to the central empirical claim of this paper. Under strict OOD settings where evaluation categories are disjoint from training categories, the two retrieval metrics tell different stories about adapter quality, and EGA is the only method that performs well on both. Table 2 reports headline numbers at $K = 1$, nprobe = 1, and Figure 4 traces Label Precision across nprobe $\in \{1, 5, 10\}$ to show that the pattern is robust to search budget. We evaluate on three benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), and Food-101 (80 seen / 21 unseen classes). For a fair comparison, all methods train identical lightweight adapters on top of frozen CLIP features and differ only in their loss functions.

On ANNS Recall, every adapter improves or matches the frozen CLIP baseline across all three OOD benchmarks (Table 2). Adapter training succeeds at *geometric* reorganization: the resulting embedding distribution is more amenable to IVF-based indexing. On Label Precision, however, the picture inverts. ICo and SRL fall *below* the frozen CLIP baseline on every dataset, with degradation reaching -32 to -36 percentage points on Food-101. The retrieved neighbors are geometrically closer in the IVF index, but they belong to the wrong semantic class. EGA is the only adapter that improves Label Precision on Aircraft (+3.6 pp) and CIFAR-10 (+0.6 pp), and on Food-101 it retains the strongest Label Precision among all adapters with bounded -8.3 pp degradation.

The Label Precision degradation of global contrastive methods persists across all three datasets and all three search budgets we tested (18 measurement points in total, Figure 4), ruling out the possibility that it is an artifact of a particular benchmark or indexing configuration. EGA’s relative advantage over global methods ranges from 4.5 pp on CIFAR-10 to 24 pp on Food-101, and tracks the difficulty of the OOD shift: when seen and unseen classes are visually similar (as in Food-101’s fine-grained dishes), the dense gradient updates of ICo and SRL pull unseen samples most aggressively into the wrong seen-class clusters.

Cross-referencing Table 1 and Table 2 reveals a structural tradeoff among adapter training paradigms. ICo and SRL achieve near-perfect Label Precision on CIFAR-100 (0.999 and 0.992) but collapse to 0.557 and 0.520 on Food-101. The frozen baseline and LoRA variants sit at the opposite extreme: modest in-distribution precision (0.55 to 0.67) but stable performance under distribution shift. EGA occupies a balanced position: 0.851 in-distribution and 0.798 on Food-101, with 0.548 on Aircraft and 0.886 on CIFAR-10. The six methods we evaluate trace out a Pareto frontier in the (ID Label

Table 2: OOD performance at $K = 1$, nprobe = 1.

Dataset	Metric	CLIP	Icon	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
CIFAR-10	AR@1	0.863	0.913	0.942	0.856	0.866	0.986
	LP@1	0.880	0.841	0.794	0.880	0.870	0.886
Aircraft	AR@1	0.773	0.875	0.863	0.911	0.881	0.898
	LP@1	0.512	0.423	0.441	0.560	0.577	0.548
Food-101	AR@1	0.842	0.811	0.827	0.902	0.899	0.886
	LP@1	0.881	0.557	0.520	0.878	0.833	0.798

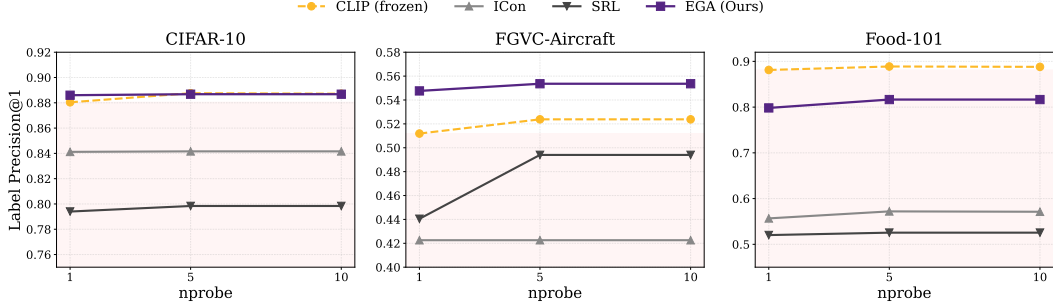


Figure 4: Label Precision@1 across three OOD benchmarks and three search budgets.

Precision, OOD Label Precision) plane, and EGA is the only adapter that is not dominated by any other: methods that exceed EGA’s in-distribution precision (Icon, SRL) collapse on OOD; methods that match or exceed EGA on OOD (LoRA, frozen) underperform substantially in-distribution.

We evaluate LoRA [10] as a parameter-efficient baseline (rank $r = 8$, $\sim 8K$ trainable parameters vs. $\sim 4.2M$ for the EGAMLP architecture used by EGA, Icon, and SRL). Two configurations are included: LoRA + InfoNCE (a global contrastive loss) and LoRA + Triplet (the same loss as EGA). Both LoRA configurations avoid the Label Precision collapse exhibited by Icon and SRL even when paired with a global loss: LoRA + InfoNCE on Food-101 reaches LP 0.878, only 0.3 pp below the frozen baseline. This suggests that the failure of Icon and SRL arises from the *combination* of (i) full-capacity adapter and (ii) dense global gradients, not from the global loss alone. With an aggressive bottleneck on adapter capacity, global losses become survivable.

4.3 Gradient Sparsity as OOD Protection

The OOD pattern in Section 4.2 raises a mechanistic question: why does EGA’s triplet objective preserve unseen-class structure while global contrastive objectives reshape it? We identify gradient sparsity at convergence as the key mechanism.

A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

As the adapter converges on seen classes, the fraction of *active* triplets decays. Figure 5 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by convergence. At steady state, 96.5% of triplets contribute zero gradient, meaning the adapter updates a sparse subset of local neighborhood relationships and leaves the remainder of the embedding space intact.

Global contrastive objectives such as Icon [1] and SRL [4] exhibit the opposite behavior. Icon minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step: every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure shapes the entire embedding distribution based on seen-class similarities, even for regions of the space occupied by unseen-class samples. The result is not loss of geometric structure but rather loss of *semantically correct* structure: unseen samples are pulled toward the clusters of seen classes that they happen to resemble in CLIP’s original metric, rather than toward other unseen samples that share their true category.

Table 3: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 4: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

Figure 5 (right) and Figure 4 together quantify the consequence: across all three OOD benchmarks, ICon and SRL fall below the frozen CLIP baseline on Label Precision (e.g., on Aircraft, ICon collapses to 0.423 from a baseline of 0.512), confirming that dense gradient updates actively harm zero-shot semantic retrieval. EGA, by contrast, improves Label Precision on Aircraft and CIFAR-10 while only 3.5% of training triplets contribute non-zero gradients at convergence.

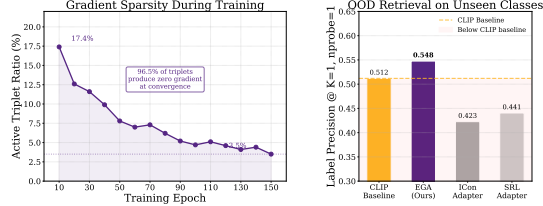


Figure 5: Gradient sparsity as OOD protection.

4.4 Ablation and Sensitivity Analysis

Table 3 isolates the contribution of each architectural component. The residual connection is the core element. Removing it causes accuracy to collapse from 0.7015 to 0.0065. This indicates that the adapter cannot reconstruct semantic relationships from target data alone and must operate as a refinement over the pre-trained features. The L2 normalization ensures the transformed embeddings remain on the unit hypersphere. Removing it decreases accuracy to 0.6590. Zero-initialization allows the adapter to begin training as an identity mapping; removing it lowers accuracy to 0.6840.

Table 4 evaluates model stability across hyperparameter variations. EGA maintains consistent retrieval quality across margin values from 0.1 to 0.3. Increasing the margin to 0.5 degrades Recall@1 to 0.7400. A larger margin forces the adapter to update embeddings that already satisfy local neighborhood constraints, which reduces gradient sparsity and alters the pre-trained manifold. Furthermore, the model shows stable performance across hidden dimensions ranging from 512 to 4096. This stability confirms that EGA achieves metric correction through its structural design rather than specific parameter capacities.

5 Conclusion

We have shown that lightweight adapter training over frozen CLIP does not produce uniformly improved retrieval embeddings, but instead lands different methods at different points along an in-distribution/out-of-distribution Pareto frontier. High-capacity adapters trained with global contrastive objectives (ICon, SRL) reach near-perfect Label Precision on CIFAR-100 (0.999, 0.992) but collapse below the frozen baseline by 32 to 36 percentage points on Food-101. Low-capacity LoRA adapters avoid this collapse but underperform substantially in-distribution. EGA, our proposed manifold-preserving adapter, occupies a balanced Pareto-optimal position: it reaches 0.851 in-distribution Label Precision while improving over the frozen baseline on two of three OOD benchmarks and bounding the degradation on the third. The mechanism is gradient sparsity at convergence, i.e., 96.5% of training triplets produce zero gradient at steady state, which permits high-capacity in-distribution refinement while leaving most of the embedding space, including regions occupied by unseen categories, largely unmodified.

The broader observation is that adapter design choices for foundation-model retrieval should be evaluated on the joint (ID, OOD) frontier rather than on either axis in isolation. Methods that compress seen-class clusters most aggressively are not necessarily the methods that generalize best, and methods that generalize best are not necessarily the methods that refine seen classes most precisely. Gradient sparsity at convergence is a measurable training-time property that predicts where an adapter will land on this frontier, and we suggest it as a useful diagnostic for future adapter design.

References

- [1] Shaden Naif Alshammari, John R Hershey, Axel Feldmann, William T Freeman, and Mark Hamilton. I-con: A unifying framework for representation learning. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- [2] Qi Chen, Bing Zhao, Haidong Wang, Mingqin Li, Chuanjie Liu, Zengzhong Li, Mao Yang, and Jingdong Wang. Spann: highly-efficient billion-scale approximate nearest neighbor search. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, NIPS ’21, Red Hook, NY, USA, 2021. Curran Associates Inc.
- [3] Haya Diwan, Jinrui Gou, Cameron Musco, Christopher Musco, and Torsten Suel. Navigable graphs for high-dimensional nearest neighbor search: Constructions and limits. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 59513–59531. Curran Associates, Inc., 2024.
- [4] Zijian Dong, Yilei Wu, Chongyao Chen, Yingtian Zou, Yichi Zhang, and Juan Helen Zhou. Improve representation for imbalanced regression through geometric constraints. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [5] Sedigheh Eslami and Gerard de Melo. Mitigate the gap: Improving cross-modal alignment in CLIP. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [6] Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics.
- [7] Hao Guo and Youyou Lu. Achieving low-latency graph-based vector search via aligning best-first search algorithm with ssd. In *19th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, pages 171–186, Boston, MA, 2025. USENIX Association.
- [8] Hao Guo and Youyou Lu. Odinann: Direct insert for consistently stable performance in billion-scale graph-based vector search. In *24th USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, CA, 2026. USENIX Association.
- [9] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2790–2799. PMLR, 09–15 Jun 2019.
- [10] Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [11] Jie Hu, Li Shen, Samuel Albanie, Gang Sun, and Enhua Wu. Squeeze-and-excitation networks. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(8):2011–2023, August 2020.
- [12] Piotr Indyk and Haike Xu. Worst-case performance of popular approximate nearest neighbor search implementations: Guarantees and limitations. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 66239–66256. Curran Associates, Inc., 2023.
- [13] Elias Jääsaari, Ville Hyvönen, and Teemu Roos. LoRANN: Low-rank matrix factorization for approximate nearest neighbor search. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [14] Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *Computer Vision – ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXIII*, page 709–727, Berlin, Heidelberg, 2022. Springer-Verlag.

- [15] Dong Jing, Xiaolong He, Yutian Luo, Nanyi Fei, Guoxing Yang, Wei Wei, Huiwen Zhao, and Zhiwu Lu. FineCLIP: Self-distilled region-based CLIP for better fine-grained understanding. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [16] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.
- [17] Kate Keahey, Jason Anderson, Zhuo Zhen, Pierre Riteau, Paul Ruth, Dan Stanzione, Mert Cevik, Jacob Collieran, Haryadi S. Gunawi, Cody Hammock, Joe Mambretti, Alexander Barnes, François Halbach, Alex Rocha, and Joe Stubbs. Lessons learned from the chameleon testbed. In *Proceedings of the 2020 USENIX Annual Technical Conference (USENIX ATC '20)*. USENIX Association, July 2020.
- [18] Hyungbin Kim, Incheol Baek, and Yon Dohn Chung. Decoupled contrastive learning for federated learning, 2025.
- [19] Panagiotis Koromilas, Efthymios Georgiou, Giorgos Bouritsas, Theodoros Giannakopoulos, Mihalis A. Nicolaou, and Yannis Panagakis. A principled framework for multi-view contrastive learning, 2025.
- [20] Chiara Lanza, Roberto Pereira, Marco Miozzo, Eduard Angelats, and Paolo Dini. Degradation of feature space in continual learning, 2026.
- [21] Fengming Lin. Vision language models: A survey of 26k papers, 2025.
- [22] Yu A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(4):824–836, April 2020.
- [23] Avik Pal, Max van Spengler, Guido Maria D’Amely di Melendugno, Alessandro Flaborea, Fabio Galasso, and Pascal Mettes. Compositional entailment learning for hyperbolic vision-language models. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [24] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, Proceedings of Machine Learning Research, pages 8748–8763. PMLR, 2021.
- [25] Sameera Ramasinghe, Violetta Shevchenko, Gil Avraham, and Ajanthan Thalaiyasingam. Accept the modality gap: An exploration in the hyperbolic space. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 27253–27262, 2024.
- [26] Simone Ricci, Niccolò Biondi, Federico Pernici, Ioannis Patras, and Alberto Del Bimbo. λ -orthogonality regularization for compatible representation learning. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026.
- [27] Daniele Savietto, Declan Campbell, André Panisson, Marco Nurisso, Giovanni Petri, Jonathan D. Cohen, and Alan Perotti. The geometry of representational failures in vision language models, 2026.
- [28] Shivam Rajendra Rai Sharma, Xiaoguang Zhu, Luca Cerny Oliveira, Kartik Patwari, La Rissa Vasquez, David Garcia, Louise Nicole C. Sevilla, Brittany N Dugger, and Chen-Nee Chuah. Benchmarking parameter efficient adaptation of vision language models on pathology. In *NeurIPS 2025 Workshop for Imageomics: Discovering Biological Knowledge from Images Using AI*, 2025.
- [29] Suhas Jayaram Subramanya, Devvrit, Rohan Kadekodi, Ravishankar Krishaswamy, and Harsha Vardhan Simhadri. Diskann: fast accurate billion-point nearest neighbor search on a single node. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, 2019.

- [30] Hangke Sui, Yuqing Wang, and Minh N. Do. Unicon: Unified framework for efficient contrastive alignment via kernels. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [31] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- [32] Liang Wang, Xiang Tao, Qiang Liu, Shu Wu, and Liang Wang. Rethinking graph masked autoencoders through alignment and uniformity. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence, AAAI’24/IAAI’24/EAAI’24*. AAAI Press, 2024.
- [33] Tongzhou Wang and Phillip Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In *Proceedings of the 37th International Conference on Machine Learning, ICML’20*. JMLR.org, 2020.
- [34] Mingqi Wu, Qiang Sun, and Archer Y. Yang. PCA++: How uniformity induces robustness to background noise in contrastive learning. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026.
- [35] Baili Xiao, Zhibin Dong, Ke Liang, Suyuan Liu, Siwei Wang, Tianrui Liu, Xingchen Hu, En Zhu, and Xinwang Liu. Easemvc: Efficient dual selection mechanism for deep multi-view clustering. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20716–20726, 2025.
- [36] Xuqian Xue, Yiming Lei, Qi Cai, Hongming Shan, and Junping Zhang. PROTOCOL: Partial optimal transport-enhanced contrastive learning for imbalanced multi-view clustering. In *Forty-second International Conference on Machine Learning*, 2025.
- [37] Ming Yang, Yuzheng Cai, and Weiguo Zheng. Cspg: Crossing sparse proximity graphs for approximate nearest neighbor search. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 103076–103100. Curran Associates, Inc., 2024.
- [38] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3813–3824, 2023.
- [39] Renrui Zhang, Wei Zhang, Rongyao Fang, Peng Gao, Kunchang Li, Jifeng Dai, Yu Qiao, and Hongsheng Li. Tip-adapter: Training-free adaption of clip for few-shot classification. In *Computer Vision – ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXV*, page 493–510, Berlin, Heidelberg, 2022. Springer-Verlag.
- [40] Yingwen Zhang, Meng Wang, Xihua Sheng, Peilin Chen, Junru Li, Li Zhang, and Shiqi Wang. An information-theoretic regularizer for lossy neural image compression. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 15573–15582, October 2025.
- [41] Qingyan Zhao, Ruifang He, Jinpeng Zhang, Chang Liu, and Bo Wang. Representation de-generation problem in prompt-based models for natural language understanding. In Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13946–13957, Torino, Italia, May 2024. ELRA and ICCL.
- [42] Xin Zhou, Yongjie Wang, and Zhiqi Shen. Cm³: Calibrating multimodal recommendation, 2025.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The paper presents both theoretical and experimental evidence to support its claims.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: We discuss the limitations of our approach in the conclusion.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Full proofs are provided in the appendix, and all assumptions are clearly stated in the main paper and/or the appendix.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: All code and data needed to reproduce the main experimental results are available on the anonymous Github repository.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All the details to reproduce the main experimental results are provided in the anonymous Github repository, including instructions for data access and preparation, and scripts to reproduce all experimental results for the new proposed method and baselines.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: All such details are provided in the anonymous Github repository, and the main paper includes a summary of the experimental setting.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The results are pretty stable so we did not report error bars; we explain this in the main paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: These details are provided in the anonymous Github repository, and the main paper includes a summary of the compute resources used for the experiments.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We believe we have followed the NeurIPS Code of Ethics in all aspects of our research.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: [N/A]

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: [N/A]

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [N/A]

Justification: [N/A]

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: [N/A]

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: [N/A]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: [N/A]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.

- 808 • We recognize that the procedures for this may vary significantly between institutions
809 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
810 guidelines for their institution.
811 • For initial submissions, do not include any information that would break anonymity (if
812 applicable), such as the institution conducting the review.

813 **16. Declaration of LLM usage**

814 Question: Does the paper describe the usage of LLMs if it is an important, original, or
815 non-standard component of the core methods in this research? Note that if the LLM is used
816 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
817 scientific rigor, or originality of the research, declaration is not required.

818 Answer: [N/A]

819 Justification: [N/A]

820 Guidelines:

- 821 • The answer [N/A] means that the core method development in this research does not
822 involve LLMs as any important, original, or non-standard components.
823 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
824 be described.

825 A Proof of Proposition 1

826 *Proof.* We prove each claim in turn.

827 **(1) Gradient support.** The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \ell_{apn}$, where
 828 $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_{\theta} \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_{\theta} \delta_{apn}, \quad (12)$$

829 which is exactly zero whenever $\delta_{apn} + m \leq 0$. Therefore, $\nabla_{\theta} \mathcal{L}$ receives contributions only from
 830 active triplets satisfying Eq. (8).

831 **(2) Convergence of $\rho(t)$.** Under mild regularity conditions (Lipschitz continuity of f_{θ} and bounded
 832 embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity,
 833 $\rho(t)$ converges to ρ^* as defined in Eq. (10), since $\rho(t)$ is a continuous functional of the embedding
 834 distribution and the embeddings converge.

835 **(3) Empirical locality (conjecture).** Strictly speaking, parameter updates $\nabla_{\theta} \delta_{apn}$ computed on
 836 seen-class triplets affect f_{θ} globally (including its behavior on unseen-class inputs) since f_{θ} is a
 837 parametric MLP and is not locally supported on the input manifold. We therefore do not claim formal
 838 zero-gradient over unseen regions. Instead, we conjecture an *empirical* locality property: when (i) the
 839 residual structure with zero initialization keeps the cumulative perturbation magnitude bounded by the
 840 sum of training gradients, and (ii) gradient sparsity at convergence ($\rho^* \approx 3.5\%$ on FGVC-Aircraft)
 841 attenuates this sum, the resulting perturbation in unseen-class regions is small enough to preserve
 842 the pretrained semantic structure for retrieval purposes. This conjecture is empirically supported
 843 by the OOD Label Precision results in Section 4.2 and Figure 4, where EGA preserves or improves
 844 Label Precision on unseen categories while ICon and SRL, which lack the gradient sparsity property,
 845 collapse below the frozen baseline. A formal Lipschitz-based bound on this perturbation is left for
 846 future work.

847 **(4) Dependence on m .** The residual distance gap δ_{apn}^* at convergence is bounded by the within-class
 848 diameter $\Delta_c = \max_{x,x' \in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c . For $m \leq \min_c \Delta_c$, the adapter can drive all
 849 within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets
 850 remain active at convergence because the within-class spread exceeds the margin, resulting in a
 851 strictly positive ρ^* . This explains the empirically observed increase in ρ^* with larger m reported in
 852 Table 4. $\square$

853 B More Experimental Details

854 **Datasets.** We evaluate on four datasets covering both in-distribution and OOD retrieval scenarios.
 855 **CIFAR-100** (100 classes, 60K images) serves as the in-distribution benchmark: the adapter is trained
 856 and evaluated on the same class label space, with a 75/25 split between database and query sets.
 857 **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained
 858 on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained
 859 aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split
 860 we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20%
 861 unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on
 862 the unseen split. Image embeddings for all datasets are extracted with frozen CLIP ViT-B/32.

863 **Baselines.** We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing
 864 the same training data, evaluation protocol, and 75/25 retrieval split:

- 865 • **ICon** [1]: a global contrastive objective minimizing the KL divergence between the empirical
 866 similarity distribution and the class-membership distribution. We instantiate ICon on the
 867 same EGAMLP architecture used by our method ($\sim 4.2\text{M}$ parameters).
- 868 • **SRL** [4]: a structural regularization objective jointly optimizing batch-wide uniformity and
 869 within-class homogeneity, also instantiated on the EGAMLP architecture.
- 870 • **LoRA + InfoNCE** [10, 31]: a low-rank adapter (rank $r = 8$, $\alpha = 16$, $\sim 8\text{K}$ parameters) with
 871 zero-initialized residual structure, trained with supervised InfoNCE, which is a standard
 872 global contrastive loss.

873 • **LoRA + Triplet** [10]: identical low-rank architecture as above, but trained with the same
874 small-margin ($m = 0.2$) triplet loss as EGA.

875 The two LoRA configurations isolate the effect of architectural capacity vs. loss function: comparing
876 LoRA + InfoNCE against ICon/SRL holds the loss family roughly fixed while changing capacity,
877 and comparing LoRA + Triplet against EGA holds the loss fixed while changing capacity.

878 **Indexing and metrics.** For all retrieval evaluations we use FAISS [16] IndexIVFFlat with
879 nlist=10 centroids and report results at $nprobe \in \{1, 5, 10\}$. We measure both Label Preci-
880 sion (LP@ K , Eq. 1) and ANNS Recall (AR@ K , Eq. 2) at $K \in \{1, 3, 5, 10\}$.

881 **Training.** All adapters are trained for 150 epochs with AdamW (weight decay 10^{-4}), cosine-
882 annealed learning rate, and batch size 256. EGAMLP-based methods (EGA, ICon, SRL) use learning
883 rate 10^{-4} ; LoRA variants use 10^{-3} to compensate for their smaller parameter count. Triplet-based
884 methods sample one positive and one negative per anchor uniformly within each mini-batch; global
885 contrastive methods (ICoN, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix.
886 All experiments use a fixed random seed (42) for reproducibility.

887 **Compute platform.** All experiments were conducted on the Chameleon Cloud [17] platform using
888 a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA
889 A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.